

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Remote Sensing and GIS L-T-P: 3-0-0 Credits:

3. Pre-requisites: none

4. Syllabus and reference books:

Syllabus:

Basic Principles of Optical Remote Sensing : Energy sources and radiation principles, Overview: basic principles of remote sensing; energy sources; electromagnetic radiation and its interactions with earth's surface and atmosphere; data acquisition; reflectance, ideal remote sensing system, different satellites programs, application of remote sensing in natural hazard study, ground water, mineral exploration, structural geology, land use and regional planning.

Aerial photography: basic components, photo interpretation elemental characters, parallax, vertical exaggeration, relief displacement, identification of different types of rocks in aerial photographs, photogrammetric techniques, world referencing system.

Digital Remote Sensing: pre-processing of satellite images -geometric rectification, atmospheric and radiometric corrections; digital image processing – spectral, spatial and radiometric enhancement techniques – contrast stretching, histogram equalization, Gaussian distribution; image arithmetic, spatial domain filtering, frequency domain filtering, image transformation-principal component analysis; supervised and unsupervised classification of satellite image for thematic map generation.

Basic GIS: Raster and Vector data, Spatial analysis, spatial interpolation techniques. Applications

Reference Books:

1) Lillesand, T.M. and Kiefer, R.W., 1987, Remote Sensing and Image Interpretation, Sec. Ed., John Wiley and Sons, Inc.: Toronto.

2) Jensen, J.R., 1986, Digital Image Processing, an Introductory Perspective. Prentice-Hall: Englewood Cliffs, N.J.

3) Richards, J.A., 1986, Digital Image Processing, Springer-Verlag: New York.

4) Remote Sensing for Geoscientists Image Analysis and Integration, Third Edition
By Gary L. Prost. CRC Press.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Basic Principles of Optical Remote Sensing	5
2.	Aerial photography:	2
3.	Digital Remote Sensing:	10
4.	Basic GIS:	3
Total number of hours		40